Kailen Swensen

Audio Software Engineer

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Audio software engineer who builds real-time C++/JUCE plugins and agentic development systems. Developed the commercial TAPE-MINI plugin, founded Lyd Audio, and runs specialized AI agents.

Skills

Audio & DSP: Real-time C++, JUCE, VST3/AU, filter design, FFT/spectral analysis, convolution, interpolation, oversampling, anti-aliasing, dynamics, delay/reverb

Agentic Systems: Claude Code, Hermes, OpenClaw, LM Studio (small local agentic models), multi-agent workflows, agent instructions and persistent memory, tool and CLI design for agents, human approval gates, automated verification, token budgeting

Languages: C++17, C, Python, JavaScript, HTML/CSS, Pure Data

Tools: CMake, Git, SVN, GitHub Actions, Linux, Reaper, Wwise, FMOD, Unreal Engine, Doxygen

Work Experience

Founder / Audio Software Developer - *Lyd Audio* *July 2026 - Present*

- Coded Bitronix, a bit-crusher and distortion plugin in C++/JUCE (in development), with an anti-aliased wavefolder measuring 7 dB less aliasing than a plain fold and real-time oscilloscope, vectorscope and 3D spectrum displays.
- Designed the concepts, aesthetics and systems for a 12-plugin line in development, including a delay with a 3-minute scrubbable history and a 16-voice synth.
- Directed Claude Code agents to implement the other plugins on a shared DSP/GUI toolkit, with repo-level instructions, a build-and-audition loop and cross-platform CI, while owning product and sound decisions.

Co-Owner / Audio Plug-in Developer - *Loft Audio Plugins* *July 2024 - September 2026*

- Handled all plugin development and Windows deployment at a three-person company, shipping TAPE-MINI (VST3, AU; on sale for macOS and Windows) with tape saturation, three-band and mid-side processing, and low-pass and high-pass filters.
- Built the UI from designer-provided assets, with thread-safe communication between the audio processor and UI.
- Added oversampling, preset management, favoriting and undo/redo, and wrote the signal flow chart and tech roadmap.

Jotform Developer (contract) - *Rixen's Enterprises* *May 2024 - November 2024*

- Built internal and customer-facing Jotform forms and apps for expenses, installation records and service scheduling.

Server/Bartender - *Kochendorfer's, Seattle* *January 2026 - Present*
Server/Bartender - *Skal Beerhall* *July 2021 - July 2026*

Projects

Designer/Developer - *Multi-Agent Workstation (Claude Code, Python, Linux)* *Feb 2026 - Present*

- Designed five agents for plugin development, job search, smart home, investing and simulation, each with its own dashboard.
- Built a festival outreach agent (Hermes, OpenClaw, Gmail/Sheets APIs) that emails an auto-updated list of verified contacts.
- Made human approval the default: agents need sign-off to submit or execute, unless a task is explicitly set up as automated.
- Built a multi-agent society simulation with decisions from an external model API, held to a $0.25/day budget.

Audio Programmer/UI Programmer - *Multi-Band Compressor (JUCE, C++, solo project)* *Sep 2022 - Sep 2023*

- Developed Linkwitz-Riley and Butterworth filter classes in C++ and a band-splitting algorithm for per-band processing.
- Used the FFT to graph frequency content before and after compression, in a real-time UI with filter and compression curves.

Audio Programmer/VFX Programmer - *Witchpunk (published on Steam, Unreal Engine, team of 27)* *Sep 2020 - Dec 2022*

- Integrated Wwise into Unreal Engine and programmed audio events and music to vary with game and player state.
- Implemented initial enemy AI with behavior trees, and worked with artists on a quick VFX pipeline in Niagara.
- Composed music for all game states and cinematic/trailer audio, and designed most of the SFX in Reaper.

Education

Bachelor of Science: Computer Science and Digital Audio - *DigiPen Institute of Technology, Redmond, WA* *April 2023*
Associate of Science: Audio Engineering Technology - *Honolulu Community College, Honolulu, HI* *May 2016*